

Heritage Treasures: An In-Depth Analysis of
UNESCO World Heritage Sites in Tableau

TEAM ID	SWTID-2026-4638
PROJECT TITLE	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture

The architecture of the "Heritage Treasures" analytics platform is structured as a three-tier framework that ensures clean separation between raw data sources, calculated analytical logic, and the interactive presentation layer. The bottom tier is the Data Ingestion and Storage Layer, which acts as the foundation of the project. This layer reads raw data from the UNESCO World Heritage registries, geographic coordinate files, and threat tracking indexes. These files are consolidated and transformed into a high-performance Tableau Hyper extract (`.hyper`), which restructures the data into an optimized columnar format to enable lightning-fast query execution across thousands of heritage records. The middle tier is the Analytical Logic and Data Modeling Layer, which processes the structured data and drives the interactive features of the platform. Using Tableau's logical relationships layer, the dataset is joined using unique site identification attributes. Within this tier, custom level-of-detail (LOD) calculations, calculated fields, and conditional flags are processed. These metrics—such as regional danger concentrations and site age benchmarks—are calculated dynamically based on the user's active filters, ensuring data consistency across all connected visualizations without altering the underlying source files. The top tier is the Visual Interaction and Presentation Layer, which provides the user interface for exploring the data. Built using the advanced dashboard tools in Tableau 2025.3.1, this layer handles all user inputs, including parameter shifts, timeline filter adjustments, and map hover actions. When a user interacts with a dashboard, the request is sent down to the Hyper engine, which returns the filtered data to update the maps, charts, and metrics instantly.